View Reviews

Paper ID
2211

Paper Title
LoRA-Based Resource-Efficient LLM Adaptation for RTL Hardware Trojan Detection

Track Name
Track 2: Semiconductors, Nanoelectronics & Next-Gen Computing

Reviewer #1

Questions

4. Please write detailed Comments for - Suitability of the title and quality of the abstract. - Adequacy of literature review and proposed methods -Quality of result analysis and conclusion

1. The authors should provide more details about the synthetic Trojan generation.
2. The authors should carefully check the manuscript for grammatical errors, typographical mistakes, and sentence construction.
3. The methodology should be explained more clearly.

Reviewer #4

Questions

4. Please write detailed Comments for - Suitability of the title and quality of the abstract. - Adequacy of literature review and proposed methods -Quality of result analysis and conclusion

The title is appropriate and accurately represents the main contribution: LoRA-based, resource-efficient LLM adaptation for RTL Hardware Trojan detection. The abstract clearly identifies the hardware-security problem, the lightweight Qwen2.5-1.5B-Instruct/LoRA approach, the synthetic dataset, the classification and localization tasks, and the principal quantitative results. The abstract is generally informative and well structured. A minor improvement would be to state more explicitly that the evaluation is based on a controlled synthetic RTL Trojan environment so that the scope of the results is immediately clear.

Adequacy of literature review and proposed methods:
The literature review adequately motivates the limitations of conventional testing, handcrafted approaches, and computationally expensive LLM adaptation. The proposed

workflow is reasonably systematic and includes RTL preprocessing, controlled Trojan insertion, obfuscation-aware prompt preparation, LoRA-based adaptation, and localization-aware evaluation. The paper also attempts to reduce shortcut learning by removing explicit Trojan markers before training and evaluation.

The main methodological limitation is the use of a synthetic controlled dataset. Strong results obtained under this setting do not yet establish generalization to heterogeneous real-world RTL designs, unseen Trojan families, or industrial benchmarks. The paper should also strengthen comparison with relevant state-of-the-art baselines and clarify whether the data split prevents structurally similar RTL modules or Trojan templates from appearing across training and testing.

Quality of result analysis and conclusion:
The paper provides quantitative classification and localization analysis and also discusses computational efficiency. The reported localization metrics and comparison of the low-resource setup with PEARL-style approaches are useful. The authors appropriately acknowledge that the present work is limited to synthetic RTL-level Trojan representations and is not yet validated for gate-level, physical-design, or large-scale industrial scenarios.

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Reviewer #6

Questions

4. Please write detailed Comments for - Suitability of the title and quality of the abstract. - Adequacy of literature review and proposed methods -Quality of result analysis and conclusion

The papers present a motivated approach for detection of hardware trojan at the RTL level.